SIMATS ENGINEERING

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I sanjay kumar soy(192311151) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: sanjay kumar soy

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. **5.** A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Answer

1.Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

A hypervisor is software that creates and manages virtual machines (VMs). There are two types of hypervisors: Type-1 and Type-2.

Type-1 Hypervisor	**Type-2 Hypervisor**
Runs directly on physical hardware	Runs on top of a host operating system
High performance with low overhead	Slower because it depends on the host OS
More secure with fewer attack points	Less secure since the host OS can be attacked
Supports centralized management and large-scale deployment	Easier for personal use but limited for enterprise environments

Analysis

- **Performance:** Type-1 hypervisors provide better speed because they communicate directly with the hardware. Type-2 hypervisors have extra overhead due to the host operating system, so they are comparatively slower.
- **Security:** Type-1 hypervisors are more secure because there is no host operating system that attackers can target. In Type-2 hypervisors, if the host OS is compromised, the virtual machines may also be affected.
- **Manageability:** Type-1 hypervisors are designed for cloud data centers and support advanced features such as live migration, high availability, and centralized management. Type-2 hypervisors are mainly used for development, testing, and learning purposes.

Justification

For a cloud datacenter hosting mission-critical applications, Type-1 Hypervisor is the better choice. It offers higher performance, stronger security, and better management features, making it reliable for applications that require high availability and minimal downtime. Type-1 hypervisors are the preferred choice for mission-critical cloud environments because they deliver better performance, improved security, and efficient management compared to Type-2 hypervisors.

2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Cloud providers often host applications for multiple organizations on the same physical server. For example, a single server may host a finance application, a healthcare system, and a student learning portal. Since these applications contain sensitive data, they must be isolated from one another to ensure security and privacy. Virtualization architecture makes this possible by creating multiple Virtual Machines (VMs) on a single physical server.

How Virtualization Supports Isolation

Virtualization uses a hypervisor to divide one physical server into multiple independent virtual machines. Each virtual machine behaves like a separate computer with its own:

- Operating System
- Applications
- CPU allocation
- Memory
- Storage
- Network settings

Although all virtual machines share the same hardware, they cannot directly access each other's data.

For example:

- VM 1 hosts the Finance Application
- VM 2 hosts the Healthcare System
- VM 3 hosts the Student Portal

If the student portal crashes or becomes infected with malware, the finance and healthcare applications continue to operate normally because each VM is isolated.

This isolation improves:

- Security – Prevents unauthorized access between applications.
- Reliability – Failure in one VM does not affect others.
- Resource Management – CPU, memory, and storage can be allocated independently.
- Cost Efficiency – Multiple applications share the same hardware instead of requiring separate servers.

Role of the Hypervisor

The hypervisor is the key component responsible for isolation.

It performs the following tasks:

- Creates and manages virtual machines.
- Allocates hardware resources to each VM.
- Prevents one VM from accessing another VM's memory or storage.
- Ensures secure communication between virtual machines and hardware.

Examples include VMware ESXi, Microsoft Hyper-V, and KVM.

Major Attack Surfaces

Although virtualization improves security, it is not completely risk-free. Two major attack surfaces are:

1. Hypervisor Attack

If an attacker successfully compromises the hypervisor:

- All virtual machines become vulnerable.
- Sensitive data from finance, healthcare, and student systems can be accessed.
- The attacker may gain complete control over the server.

Therefore, the hypervisor must be regularly updated and protected with strong security controls.

2. VM Escape Attack

VM Escape attack occurs when malicious software inside one virtual machine escapes its isolated environment and gains access to the host system or other virtual machines.

If malware enters the student Portal VM, it may attempt to access the Finance VM or healthcare VM.

This can lead to:

- Data theft
- Unauthorized access
- Privacy violations
- Service disruption

Security Measures

- Keep the hypervisor updated with security patches.
- Use strong authentication and access control.
- Enable firewalls and network segmentation between VMs.
- Encrypt sensitive data.
- Continuously monitor virtual machines for suspicious activities.
- Perform regular security audits.

3.Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

virtualization corelation VM utilisation usually gives the following recommendation :

- CPU 92%
- memory 88%
- storage I/O latency 35 ms
- average VM boot time 170 s

Analysis of the VM Host Utilization

1. CPU Utilization – 92%

The CPU is almost fully utilized.

- Most of the processor resources are already occupied.

- Virtual machines may compete for CPU time.
- Applications may respond slowly.
- Users may experience delays while running programs.

Likely bottleneck:

High CPU utilization is causing processing delays and reducing overall system performance.

2. Memory Utilization – 88%

The memory usage is very high.

- Only a small amount of RAM is available.
- The host may start using disk space as virtual memory (swapping).
- Swapping is much slower than physical RAM.

Likely Bottleneck: Insufficient free memory increases response time and affects VM performance.

3. Storage I/O Latency – 35 ms

Storage latency measures the time taken to read or write data.

- A latency of 35 ms is considered high for modern cloud environments.
- Virtual machines wait longer for storage operations.
- Applications that frequently access disks become slower.

Likely Bottleneck:

Slow storage performance delays data access and reduces VM efficiency.

4. Average VM Boot Time – 170 Seconds

Normally, a virtual machine should start much faster.

A boot time of 170 seconds (almost 3 minutes) indicates that:

- CPU resources are overloaded.
- Memory is nearly exhausted.
- Storage operations are slow.

Likely Bottleneck: The combination of high CPU usage, limited memory, and storage latency increases VM startup time.

Recommended Corrective Actions

1. Reduce CPU Load

- Distribute some virtual machines to another physical host.
- Increase CPU resources or add more processors.
- Optimize applications consuming excessive CPU.

Benefit: Improves processing speed and reduces delays.

2. Increase Memory Capacity

- Add more RAM to the host.

- Allocate memory efficiently among virtual machines.
- Remove unused or idle virtual machines.

Benefit: Reduces swapping and improves VM performance.

3. Improve Storage Performance

- Replace HDDs with SSDs or NVMe storage.
- Use faster storage controllers.
- Optimize disk I/O by removing unnecessary read/write operations.

Benefit: Reduces storage latency and improves application response time.

4. Balance Workloads

- Use load balancing to distribute VMs across multiple hosts.
- Continuously monitor resource utilization.
- Enable automatic scaling when demand increases.

Benefit: Prevents overloading of a single host.

4.Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations

Software Defined Datacenter (SDDC) is a modern datacenter where all major resource: compute, storage, and networking; are virtualized and managed through software. Unlike a traditional datacenter, which depends on manual hardware configuration, an SDDC automates resource management. This makes the datacenter more flexible, scalable, and efficient.

Traditional Datacenter vs Software Defined Datacenter

Traditional Datacenter	**Software Defined Datacenter (SDDC)**
Hardware is configured manually	Resources are managed through software
Slower deployment of services	Faster deployment and provisioning
Limited scalability	Easy and automatic scalability
Higher maintenance cost	Lower operational cost
Manual resource management	Automated and centralized management

Analysis:

In a traditional datacenter, adding a new server or storage device often requires manual installation and configuration, which takes time. In contrast, an SDDC allows administrators to provision resources in minutes using software, making it much more agile.

Virtualization Considerations

1. Compute Virtualization

Compute virtualization divides one physical server into multiple Virtual Machines (VMs) using a hypervisor.

Benefits:

- Multiple VMs can run on one server.
- New virtual machines can be created quickly.
- Hardware resources are used efficiently.
- Applications can be deployed faster.

Eg: If a company needs a new application server, an administrator can create a VM within minutes instead of purchasing and configuring a new physical server.

2. Storage Virtualization

Storage virtualization combines multiple physical storage devices into a single virtual storage pool.

Benefits:

- Storage can be allocated dynamically.
- Easier backup and recovery.
- Better storage utilization.
- Faster access to storage resources.

Eg: If one application requires more storage, additional space can be assigned through software without replacing physical disks.

3. Network Virtualization

Network virtualization creates virtual networks using software instead of relying only on physical switches and routers.

Benefits:

- Faster network configuration.
- Better network security through virtual isolation.
- Easier management of network resources.
- Supports rapid deployment of new applications.

Eg: Different departments such as Finance, HR, and Student Services can have separate virtual networks while sharing the same physical infrastructure.

how sddc improves agility

An SDDC improves agility in several ways:

- Resources can be provisioned in minutes instead of days.
- Applications can scale automatically based on demand.
- Centralized management reduces manual effort.
- Automation minimizes configuration errors.

- Organizations can respond quickly to changing business needs.

This enables faster deployment of services and improves overall productivity.

advantages of sddc

- Faster deployment of applications.
- Better utilization of hardware resources.
- Lower infrastructure and maintenance costs.
- High scalability and flexibility.
- Improved disaster recovery and business continuity.
- Simplified management through automation.

5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

multi-cloud federation is an environment where multiple cloud providers work together to deliver services. Users should be able to access resources across different cloud platforms using a single identity. However, if each cloud provider uses a different identity management system, users may face authentication problems, leading to identity mismatches.

analysis of the issue

In a multi-cloud federation, each cloud provider may use different methods to store user accounts and verify identities.

For example:

- Cloud Provider A stores the user as student123.
- Cloud Provider B stores the same user as student_123.

Although both accounts belong to the same person, the systems treat them as different users. As a result:

- Users may not be able to log in.
- Access to cloud resources may be denied.
- Duplicate user accounts may be created.
- Administrators face difficulty managing permissions.
- Security and privacy risks increase because user identities are inconsistent.

Secure Identity and Privacy Design

1. Identity Federation

Use a common Identity Provider (IdP) that is trusted by all cloud providers.

Instead of maintaining separate user accounts in every cloud, users authenticate once with the Identity Provider.

Benefit:

- One identity is recognized across all cloud platforms.
- Reduces duplicate user accounts.

2. Single Sign-On (SSO)

- Users log in only once.
- They can access multiple cloud services without entering their credentials again.

Benefit:

- Better user experience.
- Reduced password management.
- Improved security.

3. Standard Authentication Protocols

- saml (security assertion markup language)
- OAuth 2.0
- OpenID Connect (OIDC)

These protocols securely exchange authentication and authorization information between cloud providers.

Benefit:

- Different cloud providers can trust and verify the same user identity.
- Improves interoperability between cloud platforms.

4. Privacy Protection

User privacy should also be protected. This can be achieved by:

- Sharing only the minimum user information required.
- Encrypting user data during storage and transmission.
- Using secure communication protocols such as https/tls.
- Preventing unauthorized access to personal information.

Benefit:

- Protects confidential user data.
- Reduces the risk of privacy breaches.

5. Role-Based Access Control (RBAC)

Implement Role-Based Access Control (RBAC).Different users receive different permissions based on their roles.

example:

- Admin – Full access
- Faculty – Manage courses
- Student – View learning materials and submit assignments

Benefit:

- Users access only the resources they are authorized to use.
- Improves security and simplifies administration.